



ENGO 545 – Fall 2026
Hydrographic Surveying
Laboratory exercise #2 (part 1)
Speed of sound in water



Deadline: Mon, Oct 19, 2026 (11:59 PM)

Learning objectives:

After completing this laboratory exercise students should be able to:

- Calculate speed of sound in water from measurements of temperature, pressure (or depth), and salinity
- Differentiate between speed of sound profiles of constant velocity vs. constant gradient
- Estimate depth given observed two-way travel time and a speed of sound profile
- Program an algorithm in MATLAB, Python, or C/C++
- Write a technical report

Background information:

You are conducting an offshore single-beam echo sounder (SBES) survey in the Gulf of Mexico. Temperature and salinity data are available from a nearby Argo profiling float (see Table 1). Your echo sounder measures the two-way travel time (TWTT) for an acoustic signal reflected from the seabed. Example returns are 0.22 s, 0.43 s, 0.56 s, 0.67 s, and 0.86 s. The transducer is mounted 3.2 m below the water line.

Table 1. Gulf of Mexico temperature and salinity data

Depth [m]	Temperature [C°]	Salinity [ppt]
0	31.0	35.70
10	31.1	35.94
20	30.2	36.25
30	29.7	36.33
50	26.4	36.59
75	21.7	36.50
100	20.2	36.48
125	18.9	36.43
150	17.8	36.33
200	15.9	36.16
250	14.5	35.96
300	13.1	35.61
400	10.9	35.35
500	9.2	35.14
600	7.8	35.02
700	6.9	34.95
800	6.0	34.93
900	5.6	34.91
1000	5.2	34.91
1100	4.7	34.91

Tasks:

1. Given the data in Table 1, generate temperature vs. depth and salinity vs. depth plots.
 2. Compute the appropriate speeds of sound.
 3. Plot the speed of sound profile as a function of depth. This is also sometimes referred to an SVP or a sound velocity profile. Specify your choice for the type of profile (e.g., profile of constant velocity vs. constant gradient).
 4. Estimate the surveyed depths as accurately as possible. Explain your algorithm in detail, and state any assumptions.
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Deliverables:

Check the Additional Course Information document on D2L for the required deliverables on a laboratory exercise.